U.S. Trade Policy Uncertainty and Chilean Peso Repricing

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Abstract

We ask whether U.S. trade policy uncertainty (TPU) moves the Chilean peso through a risk-compensation channel, which predicts immediate depreciation together with higher terminal currency risk. Using monthly local projections on 258 common news months from 2003 to 2024, we find no evidence of immediate repricing: a one-standard-deviation TPU innovation is associated with news-month depreciation of -0.012 log percentage points (95% simultaneous interval $[-0.495, 0.470]$). The variance of the three-month terminal dollar value changes by 2.1% ($[-27.9\%, 44.6\%]$), so the joint price–risk prediction is not met, although the data cannot rule out sizable risk increases. Survey-implied forecast changes are small, with simultaneous intervals inside ± 0.5 percentage points. The peso does weaken over the following three months, by 0.63 log percentage points, but this delayed response is imprecisely estimated (simultaneous interval $[-0.21, 1.46]$) and falls by two-thirds when 2018–2019 is excluded. Realized and implied volatility, additional option tenors, and risk reversals show no stable increase in risk.

1 Introduction

Trade-policy news can weaken a currency through several economically distinct forces. Investors may expect lower future currency payoffs, require more compensation for bearing risk, or revise their assessment of interest rates and financial conditions. These explanations can generate the same direction of exchange-rate movement. This paper asks whether U.S. trade policy uncertainty (TPU) is associated with Chilean peso repricing and whether joint evidence on prices, terminal currency risk, and reported expectations supports a risk-compensation interpretation.

A two-date portfolio model makes the empirical restrictions explicit. With positive aggregate peso exposure, a mean-preserving increase in terminal dollar-value dispersion lowers the current dollar price of the peso and raises expected dollar compensation, holding interest rates and risk tolerance fixed. The prediction is static: it concerns the current price and the physical distribution of a specified terminal payoff. A depreciation several months later, a rise in realized path volatility, or an unchanged survey median would each speak to only part of it.

Chile permits a focused comparison of these objects using a long reference-rate history, market quotes, and exchange-rate surveys. We combine the newspaper TPU measure of Caldara et al. (2020) with monthly local projections over January 2003–September 2024. Our TPU innovations are residuals from a projection on lagged domestic and global variables, not pure uncertainty shocks: trade-policy coverage may also carry news about future fundamentals, and contemporaneous global surprises may move both TPU and the peso. We therefore interpret the estimates as conditional innovation responses, not causal effects.

Point estimates suggest delayed rather than immediate depreciation. The news-month response to a one-standard-deviation TPU innovation is essentially zero, while cumulative depreciation reaches 0.63 log percentage points by the third subsequent month. This delayed response is imprecise: simultaneous intervals over the primary price horizons include zero, and excluding windows that overlap 2018–2019 cuts the three-month estimate to 0.20. Because the model's prediction concerns the current price, a later cumulative movement cannot confirm it.

The main test places current depreciation, the log variance of normalized terminal dollar value, and two survey statistics on the same 258 three-month origins and common bootstrap paths. Neither price nor risk shows the response the mechanism requires. Current depreciation is -0.012 log percentage points, with simultaneous interval $[-0.495, 0.470]$, and terminal variance changes by 2.1%, with interval $[-27.9\%, 44.6\%]$. The variance interval is wide enough to admit substantial risk increases, so the test is uninformative about moderate risk effects. Survey-implied forecast changes are small, with simultaneous intervals inside ± 0.5 normalized percentage points; these are point medians reported after the news month, not investors' contemporaneous conditional expectations.

Supplementary measurements do not change this picture. Implied volatility, additional option tenors, and forward realized risk show no common, precise increase. Survey forecasts outperform two historical price rules on absolute and squared error, but their advantage does not vary detectably with TPU. We could not assemble matched investable returns, so the compensation prediction itself remains untested. The contribution is to test the price, risk, and expectations predictions of a risk-compensation channel jointly, on the same dates and within one confidence family. Doing so separates two patterns that a price regression alone would blur: TPU innovations are followed by a delayed, imprecisely estimated depreciation, but there is no evidence of the immediate repricing and higher terminal risk that the channel requires.

2 Related literature

The paper connects trade-policy uncertainty to currency pricing. Caldara et al. (2020) construct measures from newspapers, earnings calls, and tariff information and document effects on investment. Their analysis distinguishes news about future tariff levels from uncertainty about those levels. We use their newspaper measure to study the peso and treat its mapping into the physical variance of terminal currency value as an empirical question.

The identification issue also connects to Ludvigson et al. (2021), who distinguish uncertainty as an endogenous response from uncertainty as an economic impulse. In this application, common trade and global financial news can affect both the newspaper index and currency prices; conditioning on historical controls does not eliminate that problem.

The pricing framework draws on the portfolio perspective emphasized by Gabaix and Maggiori (2015), in which financial risk bearing and currency exposures affect exchange rates and expected returns. Our two-state exponential-utility model borrows this perspective without their intermediary constraints or general-equilibrium structure. It makes explicit the restrictions on aggregate exposure, expected terminal value, and risk tolerance needed to interpret depreciation as compensation for additional risk, and these restrictions, not the sign of the exchange-rate response, organize the empirical tests.

The estimation strategy follows the horizon-specific regressions of Jordà (2005). They accommodate distinct price, risk, and survey outcomes without imposing one dynamic system on all measurements. Montiel Olea and Plagborg-Møller (2021) provide persistence-robust inference for lag-augmented local projections, which motivates one of our sensitivity checks (Section 5). Plagborg-Møller and Wolf (2021) show that LPs and VARs share impulse-response estimands with

unrestricted lags. Changing estimators therefore cannot by itself identify the underlying news.

3 A conditional pricing framework

The portfolio perspective links currency prices to exposures and risk bearing (Gabaix and Maggiori, 2015). The following two-state model makes explicit the restrictions needed for a risk-compensation interpretation. It is illustrative and is not estimated.

Let Q_0 and Q_1 denote dollars per peso, and $E_0 = 1/Q_0$ the usual CLP/USD quotation. A peso asset costs Q_0 dollars and pays $R > 0$ pesos at maturity; the dollar asset pays gross return $R^* > 0$. Investor i has exponential utility $-\exp(-\gamma_i W_i)$ over terminal dollar wealth, unrestricted positions, and a background payment of Rh_i pesos. The peso asset has fixed net supply $B > 0$, whereas the dollar asset is elastically supplied. Define aggregate risk tolerance and net peso exposure as

$$T = \sum_i \gamma_i^{-1}, \qquad H = \sum_i h_i, \qquad K = B + H. \tag{1}$$

Neither a country's net external position nor a subset of derivatives holdings directly measures K.

Terminal currency value satisfies

$$Q_1 = \mu(u) + \sigma(u)\varepsilon, \qquad \Pr(\varepsilon = 1) = \Pr(\varepsilon = -1) = \tfrac{1}{2}, \qquad 0 < \sigma(u) < \mu(u). \tag{2}$$

The pure-risk experiment holds μ fixed while increasing σ. Whether a TPU innovation resembles this experiment is an empirical question.

Write $a = R\mu - R^*Q_0$ and $d = R\sigma$. Up to terms independent of asset holdings, investor i maximizes

$$n_i a - \gamma_i^{-1} \log\cosh[\gamma_i d(n_i + h_i)]\,. \tag{3}$$

For $|a| < d$, strict concavity gives demand $n_i = (\gamma_i d)^{-1} \operatorname{artanh}(a/d) - h_i$. Summing demands and imposing market clearing yields

$$Q_0 = \frac{R}{R^*}\left[\mu - \sigma \tanh\left(\frac{KR\sigma}{T}\right)\right], \qquad n_i = \frac{K}{\gamma_i T} - h_i. \tag{4}$$

This unique price is positive and strictly between the discounted state payoffs.

With $z = KR\sigma/T$, holding rates, exposure, and risk tolerance fixed gives

$$\frac{dQ_0}{du} = \frac{R}{R^*}\left[\mu'(u) - \sigma'(u)\{\tanh z + z \operatorname{sech}^2 z\}\right]. \tag{5}$$

Thus, a mean-preserving dispersion increase depreciates the peso when $K > 0$; the sign reverses when $K < 0$. Expected excess dollar payoff per asset is $\rho = R\mu - R^*Q_0 = R\sigma \tanh z$; expected simple excess return is ρ/Q_0. Both rise under the positive-exposure pure-risk experiment. Lower expected value or lower risk tolerance can also cause depreciation, however. Unchanged equilibrium holdings under fixed supplies do not imply unchanged national financial-account flows, and this two-date model supplies no dynamic response path.

The empirical questions are consequently separate: whether TPU innovations predict greater relevant risk (H1), whether the peso depreciates (H2), and whether matched expected simple excess returns increase (H3). Establishing the proposed mechanism additionally requires restrictions on expected value, rates, exposure, and risk bearing. Price direction alone cannot supply those restrictions.

4 Data and measurement

Trade policy uncertainty. The baseline uses the monthly U.S. newspaper index of Caldara et al. (2020), with the November 3, 2026 download fixed across the baseline analysis (Caldara et al., 2026). The index measures the frequency of news coverage that jointly discusses trade policy and uncertainty. It can rise when news changes the expected level of tariffs, their dispersion, or both. Observations are assigned to the month covered by the articles, so a month's index value may not have been published by that month's last quote. The sample spans January 2003–September 2024 (261 months), with earlier observations used to initialize lags. Figure 1 shows the standardized innovation and the bilateral exchange rate.

Reference exchange rate and realized risk. Let E denote Chilean pesos per U.S. dollar and $Q = 1/E$ denote U.S. dollars per peso. The Central Bank of Chile's *Dólar Observado* is a transaction-weighted reference rate for the preceding banking day (Banco Central de Chile, n.d.a). The data retain the official effective date and a separately inferred trading date, obtained from the preceding nonmissing effective day on the official calendar. The monthly endpoint E_t is the last observation assigned to trading month t. The September 2024 endpoint, for example, is assigned to September 30 trading and became effective October 1.

Let $r_d = \log E_d - \log E_{d^-}$ be the change from the previous valid daily quote. The baseline annualized realized variance proxy is

$$RV_t = \frac{252}{n_t} \sum_{d \in t} r_d^2,$$

where n_t counts valid returns by the month of their right endpoint. A return that crosses a month boundary is assigned to the new month. The series requires at least 15 observations and has 17–23 returns per baseline month. No zero returns are inserted for holidays. This measure describes the realized path of a reference price that averages transactions; it can differ from a realized measure based on market quotes and from a conditional variance of terminal dollar value.

Market and survey measurements. The market extension uses LSEG spot midquotes and the three-month USD/CLP ATM implied volatility series `CLP3MO=` (LSEG, 2026). The IV archive contains 5,594 dated bid–ask observations from July 17, 2003 through December 31, 2024, spanning 258 months. Monthly IV takes the last valid daily midpoint and enters as $2\log(\text{IV})$. A fixed unit conversion only changes the regression intercept. The export does not document the quote time, ATM strike convention, or option expiry. A separate Bloomberg export adds volatility quotes labeled 1, 2, 6, and 12 months and a signed three-month 25-delta risk reversal (Bloomberg, 2026). Its 33,413 dated records extend through November 4, 2026; the historical response regressions retain the September 2024 outcome cutoff. We keep midpoints and last quotes separate. This export also does not document quote conventions, and the currency direction of the risk reversal is unverified.

The Central Bank of Chile's Economic Expectations Survey (EEE) supplies monthly point forecasts of the CLP/USD exchange rate at horizons of 2, 11, and 23 months from the survey month (Banco Central de Chile, n.d.b). The analysis uses published medians and a separate historical release of individual responses. Reported point forecasts and cross-respondent dispersion measure different objects from the probability mean and variance faced by investors. The survey's month label, the date of the applicable reference rate, and individual response timestamps generally differ. The financial-operator survey (EOF) and indicative Comder marks provide supplementary measures with separate timing and contract limitations.

Controls and vintage. The baseline includes the VIX, the BIS broad dollar index, the IMF copper price, and the effective federal funds rate, collected via FRED (Federal Reserve Bank of St. Louis, n.d.), as well as the Chilean monetary policy rate and year-over-year IMACEC growth from the Central Bank of Chile. The lags are defined using economic periods. Because these series are revised, they do not reproduce investors' real-time information set.

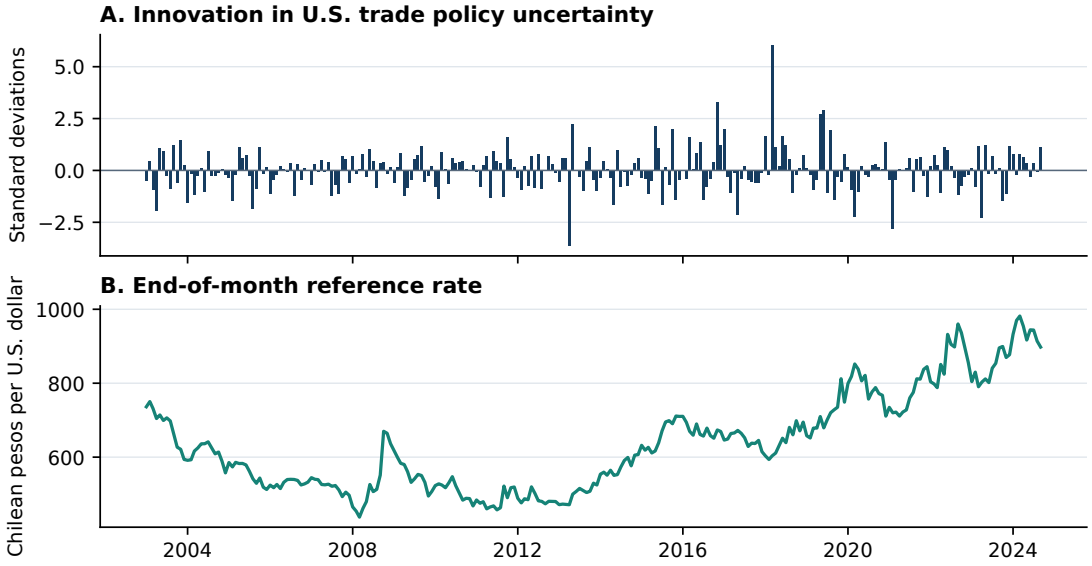


Figure 1: Monthly TPU innovations and the Chilean peso

Notes: January 2003–September 2024. The top panel uses standardized residuals from the baseline TPU projection. The lower panel uses the last Central Bank reference rate assigned to each trading month; an increase denotes peso depreciation against the dollar.

5 Empirical design and inference

5.1 Statistical innovations and local projections

Let $U_t = \log(1 + \mathrm{TPU}_t)$, using the news measure of Caldara et al. (2020). The baseline information vector W_t contains three monthly lags each of U_t, $100\Delta \log E_t$, and $q_t = \log RV_t$; the log monthly mean VIX at $t-1$, broad-dollar and copper log changes and the monthly federal funds rate at $t-2$, Chilean policy rate at $t-1$, and year-over-year IMACEC growth at $t-3$; and $\log E_{t-1}$. These are 16 nonconstant regressors. We estimate

$$U_t = a + \Pi' W_t + \xi_t, \qquad \eta_t = \widehat{\xi}_t / \widehat{s}, \qquad \widehat{s} = \left[\frac{\sum_{t=1}^{N} \widehat{\xi}_t^2}{N-1} \right]^{1/2}. \tag{6}$$

The stage uses all 261 baseline months and has $R^2 = 0.8249$ and $\widehat{s} = 0.25253$. The same stage and normalization apply across horizons.

Following Jordà (2005), separate local projections estimate

$$100[\log E_{t+h} - \log E_{t-1}] = \alpha_h + \beta_h \eta_t + \Gamma_h' W_t + e_{t,h}, \tag{7}$$

$$q_{t+h} = \delta_h + \theta_h \eta_t + \Lambda_h' W_t + v_{t,h}. \tag{8}$$

Positive β_h denotes depreciation in log percentage points per residual standard deviation. The price window includes month t: $h = 0$ is the news-month change, and $h = 3$ ends three months after

that month. Neither is an instantaneous announcement response. The baseline analysis designated $h = 0, 1, 3$ as primary horizons, with $h = 1$ the primary test for reference-rate risk. The main mechanism test, added after the baseline analysis, fixes the terminal month at $t + 3$ and uses 258 origins common to the price, model-based terminal-variance, and survey outcomes (Section 6). Longer horizons and path-risk measures are supplementary. Targets after September 2024 are excluded. The variance percentage change associated with θ_h is $100(e^{\theta_h} - 1)$; the corresponding standard-deviation change is $100(e^{\theta_h/2} - 1)$.

Because both stages contain the same complete W_t, the Frisch–Waugh–Lovell theorem implies that the coefficient on η_t equals $\widehat{s}$ times the coefficient on U_t conditional on W_t. Fixed changes to the first-stage projection coefficients therefore cancel from these linear LPs. This algebra does not make the innovation exogenous, and the estimated scale adds sampling uncertainty, which the bootstrap below accounts for by reestimating it in each draw. A causal reading would additionally require that no omitted contemporaneous news drives both measured TPU and the outcome.

5.2 Uncertainty and sensitivity

Pointwise inference uses Bartlett Newey–West covariance estimates (Newey and West, 1987), bandwidth $\max(3, h+1)$, and Student-t critical values with residual degrees of freedom. Gapped samples retain actual calendar distances. Fixed bandwidths six and twelve are separate sensitivity checks.

For the primary horizons, a circular moving-block bootstrap follows the time-series resampling principle of Künsch (1989). Each draw jointly resamples already aligned $(U_t, W_t, Y_{t,h})$ tuples, reestimates the innovation equation, and recomputes its sample standard deviation. Leads and lags are not reconstructed across artificial block joins. We use 9,999 draws for each block length (12 and 24 months); none fails. We report basic pointwise intervals and simultaneous bands from the maximum absolute coefficient deviation standardized by its bootstrap standard deviation. Price horizons form one three-coefficient family and risk horizons another; coverage is stated separately for each family, not jointly across all six coefficients. In a simulation from a simplified, highly persistent process, nominal 95% bands cover all six true responses on only about 87% (12-month blocks) and 83% (24-month blocks) of paths (Appendix G), so the reported bands may be too narrow.

Sensitivity analyses use six lags, reduced controls, alternative transformations and price aggregation, an extended sample, and episode exclusions. Episode exclusions measure how much a result depends on particular periods, and changing the innovation equation can change both the conditioning set and the units of the innovation. Lag augmentation follows Montiel Olea and Plagborg-Møller (2021), although their guarantees do not directly cover our generated innovation scale, simultaneous bands, conditional-variance estimates, or survey comparisons.

Most analyses beyond the baseline, including the main joint test, were designed after earlier results had been seen. Multiplicity adjustments within each analysis do not account for this broader search, so we treat these analyses as exploratory and report all of them, including null results and numerical failures.

6 Main evidence: prices, terminal risk, and survey-implied adjustment

Table 1 reports the main mechanism test. Every row uses the same 258 news months, January 2003–June 2024: the months for which current depreciation, three-month terminal conditional variance, and both survey quantities are available. The TPU innovation and its standard deviation

keep their 261-month baseline definitions, so differences across rows do not reflect different dates or shock scales.

6.1 Aligned economic objects

The contemporaneous price and terminal-value outcomes are

$$d_{0,t} = 100(\log E_t - \log E_{t-1}), \qquad Y_{t,3} = 100(Q_{t+3}/Q_{t-1} - 1),$$

where $Q = 1/E$ is dollars per peso. Its working conditional mean is linear in the standardized TPU innovation and W_t; log conditional variance is linear in the innovation, lagged log realized variance, and lagged log VIX. Let γ_3 denote the innovation coefficient in that log-variance equation. Conditional on the pre-news price, the proportional variance response carries over to terminal dollar value if the specified conditional moments are correct. It is a model-based estimate, not an observed variance.

For survey month $t+1$, let $\widetilde{Q}_{t+1,3}$ be the reciprocal of the median positive individual response to the EEE two-month CLP/USD question, whose target is the end of $t+3$. Let Q_{t+1}^{pre} be the latest reference price with a legal effective date strictly before the survey month's first day. All positive answers are retained. The two survey outcomes are

$$f_t^{\text{post}} = 100\,\frac{\widetilde{Q}_{t+1,3} - Q_{t+1}^{\text{pre}}}{Q_{t-1}}, \tag{9}$$

$$f_t^{\text{total}} = 100\left(\frac{\widetilde{Q}_{t+1,3}}{Q_{t-1}} - 1\right). \tag{10}$$

They express the forecast's remaining change relative to a pre-survey reference price and its total change relative to the pre-news anchor. Both use the same denominator and target. A negative coefficient indicates a weaker forecast dollar value of the peso. Neither quantity is a revision of a forecast for a fixed target. Because the survey is taken after the news month, its median is also not the model's contemporaneous arithmetic conditional mean.

6.2 Joint evidence and its precision

Panel A combines the four estimates in a single 95% maximum-statistic confidence family, using 999 common circular-block draws. The current-depreciation coefficient is -0.0124 log percentage points, with interval $[-0.4949, 0.4701]$. The terminal log-variance coefficient is 0.0206, implying a 2.08% variance increase with an interval of $[-27.91\%, 44.56\%]$. Neither response is distinguishable from zero. The variance interval is wide, however, spanning a 28% decrease to a 45% increase, so the data are uninformative about moderate risk effects.

The survey coefficients are -0.1857 and -0.1066 normalized dollar-value percentage points. Their intervals are $[-0.3973, 0.0260]$ and $[-0.4997, 0.2866]$. Panel C applies illustrative bounds of ± 0.5, ± 1, and ± 2 percentage points. Writing L_j and U_j for the simultaneous lower and upper limits, the criterion requires

$$L_{\gamma_3} > 0, \quad L_{d_0} > 0, \quad [L_{f^{\text{post}}}, U_{f^{\text{post}}}] \subset (-\delta, \delta), \quad [L_{f^{\text{total}}}, U_{f^{\text{total}}}] \subset (-\delta, \delta).$$

At every bound both survey intervals lie inside the range, but neither the variance nor the depreciation lower limit is positive, so the joint criterion fails. The survey result at $\delta = 0.5$ is marginal: the unrounded lower limit for the total-change slope is -0.4997409. The bounds are illustrative, not a calibrated economic equivalence threshold.

Table 1: Joint price–risk–survey evidence: 258 common news months

Outcome	Coefficient	Economic unit	Effect	95% simultaneous CI
A. Contemporaneous pricing and the three-month terminal object				
News-month depreciation	−0.0124	Log pp	−0.0124	[−0.4949, 0.4701]
Terminal conditional variance	0.0206	% variance	2.08	[−27.91, 44.56]
Survey-implied remaining change	−0.1857	USD-value pp	−0.1857	[−0.3973, 0.0260]
Survey-implied total change	−0.1066	USD-value pp	−0.1066	[−0.4997, 0.2866]
B. Subsequent price path: reported under a separate confidence family				
Cumulative depreciation through $t+3$	0.6263	Log pp	0.6263	[−0.2112, 1.4639]

C. Joint band-based criterion using Panel A only

Survey bound δ	Variance lower > 0	Depreciation lower > 0	Both survey bands inside	Joint criterion
0.5 pp	No	No	Yes	No
1.0 pp	No	No	Yes	No
2.0 pp	No	No	Yes	No

Notes: All panels use the identical 258 origins, January 2003–June 2024, and the original 261-month TPU standard deviation. Panel A is a single four-margin maximum-statistic family: 999 paired 12-month circular-block draws reestimate the first stage and scale. The variance coefficient is in log variance, with log interval [−0.3273, 0.3685]; its economic effect and interval use $100(e^{\gamma}-1)$. Other effects retain their coefficient units. Survey quantities are simple dollar-per-peso forecast changes normalized by the pre-news price; they are not consecutive-survey revisions. Panel B uses the original price-family 95% simultaneous band over $h = 0, 1, 3$ (9,999 draws, blocks of 12); the block-24 interval is [−0.2373, 1.4900]. Panel B is excluded from Panel A's four-margin criterion. Panel C applies a strict interval-inclusion rule to unrounded limits: both survey bands must lie inside $(-\delta, \delta)$.

Panel B reports three-month cumulative depreciation on the same 258 dates: 0.6263 log percentage points, with simultaneous interval $[-0.2112, 1.4639]$ from the baseline price family over $h = 0, 1, 3$. Its window runs from the end of $t-1$ through $t+3$, so it measures how the peso moves over the news month and the following three months, not the contemporaneous repricing the static mechanism predicts. It is therefore not part of the joint criterion.

7 Bilateral price responses

The baseline horizon profile shows little movement in the news month and positive cumulative depreciation over the following months. Appendix Table A.1 reports the three original horizons, while Figure 2 displays the full path. The $h = 3$ estimate is 0.626 log percentage points per standard deviation. Its pointwise HAC interval is $[-0.035, 1.288]$ ($p = 0.063$). The 12- and 24-month block bootstrap basic intervals are $[0.035, 1.405]$ and $[0.079, 1.459]$, respectively. When the three original price horizons are considered together, the corresponding intervals are $[-0.211, 1.464]$ and $[-0.237, 1.490]$. The pointwise bootstrap intervals exclude zero, but the family-wise intervals do not. At the supplementary horizon $h = 6$, the estimate is 0.960 (pointwise HAC $p = 0.015$).

The $h = 3$ estimate is stable across alternative controls, transformations and endpoints, ranging from 0.55 to 0.66 (Appendix A), and is essentially unchanged (0.63) in a sample extended through July 2026 with a reestimated innovation. It depends heavily on one episode, however: dropping outcome windows that overlap 2018–2019 and reestimating the first stage reduces it to 0.20 (Appendix Table A.2). The delayed depreciation is thus concentrated in the 2018–2019 trade

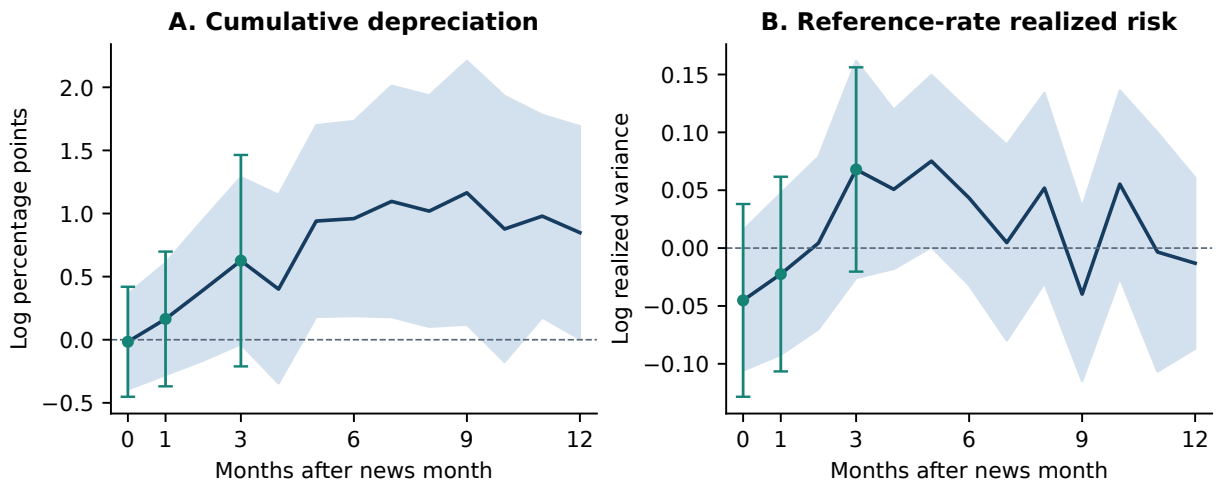


Figure 2: Local projections of depreciation and reference-rate realized risk

Notes: Shading shows pointwise 95% HAC intervals at all horizons. Capped intervals at $h = 0, 1, 3$ are 95% simultaneous bootstrap intervals computed separately for each panel using 12-month blocks. Both panels use the baseline innovation scale.

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Using common dates from LSEG spot and the Central Bank reference series gives a similar timing pattern. On 258 common origins, the $h = 3$ cumulative estimates can be decomposed exactly into a change during the news month and a change from the end of that month to $t + 3$. The contemporaneous estimates are -0.017 and -0.012 log points for the LSEG and reference series, respectively; the subsequent estimates are 0.678 and 0.639. All six simultaneous intervals include zero. The point estimates fit a delayed association; monthly date labels cannot resolve intraday repricing, and the delay need not be causal.

Evidence that the response is specific to Chile is limited: effective-rate and peer-currency comparisons are weaker. The result also depends on the news measure. The Baker–Bloom–Davis trade-policy measure yields an $h = 3$ estimate of only 0.074 log percentage points (pointwise HAC $p = 0.853$), and after residualizing on the same controls the two measures correlate only about 0.43, so they capture partly different variation. The remaining results use the Caldara index and baseline scale unless stated otherwise.

8 Supplementary risk measurements

8.1 Realized path risk and three-month implied volatility

Daily realized variation and option-implied volatility measure different objects from the terminal conditional variance in Table 1. The baseline reference-rate realized-variance response at $h = 1$ is -0.0225 (pointwise HAC $p = 0.526$). Because this measure tracks squared movements along a transaction-averaged daily path, a null here says little about the variance of the terminal payoff.

Table 2 uses LSEG `CLP3MO=` as a supplementary indicator. At response month $m = t+h$, Panel A pairs the final three-month IV midpoint with spot returns in the next three complete calendar months. Both endpoints of each adjacent spot return must lie within that window, with at least 15 segments ending in each month. The realized measure is the annualized mean squared log return. The IV and realized outcomes use identical origins at each horizon, and the calendar windows only approximate the option expiry.

Table 2: Implied risk, forward realized risk, and cumulative depreciation

h	N	Outcome 1: estimate [95% CI]	Outcome 2: estimate [95% CI]
A. Log squared IV and log forward realized risk			
0	252	-0.021 $[-0.076,\ 0.034]$	0.022 $[-0.034,\ 0.077]$
1	251	-0.022 $[-0.079,\ 0.034]$	0.034 $[-0.028,\ 0.096]$
3	249	0.021 $[-0.027,\ 0.070]$	0.048 $[-0.036,\ 0.132]$
B. Cumulative depreciation and log forward realized risk			
0	252	-0.041 $[-0.604,\ 0.522]$	0.022 $[-0.035,\ 0.078]$
1	251	0.171 $[-0.493,\ 0.834]$	0.034 $[-0.029,\ 0.098]$
3	249	0.664 $[-0.317,\ 1.645]$	0.048 $[-0.038,\ 0.135]$

Notes: Each panel is a separate six-margin family using calendar-indexed scores, Bartlett bandwidth 12 and 9,999 common Gaussian draws. Within each horizon the two outcomes share the same origins. At $m = t + h$, IV uses a three-month quote in month m and realized risk uses the three complete months after m. Depreciation is cumulative from the end of $t - 1$ through m and is measured in log percentage points.

IV coefficients are negative at $h = 0, 1$ and positive at $h = 3$; forward realized-risk coefficients are positive throughout. All six simultaneous intervals include zero under both bandwidths. Panel B pairs cumulative depreciation with forward realized risk on the same dates. At $h = 3$, the coefficients are 0.664 log percentage points and 0.048 log variance; their simultaneous intervals are $[-0.317, 1.645]$ and $[-0.038, 0.135]$. The latter risk window begins in $t + 4$ and ends in $t + 6$, unlike the fixed $t + 3$ endpoint in the main table. The pairing thus provides no evidence of concurrent positive price and risk responses; it is not part of the main criterion.

8.2 Additional option tenors and signed risk reversals

New Bloomberg data add volatility quotes labeled one, two, six and twelve months and a signed three-month risk reversal. Appendix Table E.1 reports midpoint level responses at all three original response horizons. Their simultaneous intervals include zero, as do the signed risk-reversal responses. Because the risk reversals are in supplier quote units with unverified currency direction and delta convention, their signs cannot be read as peso downside skew.

A common-date family combining spot, the five volatility tenors including LSEG's three-month series, and the risk reversal contains no interval that excludes zero. None of its fifteen positive price–IV lower-bound checks is met. Further comparisons pair the four new tenors with subsequent calendar-path realized risk of approximate matching length; all 36 simultaneous intervals contain zero. These results provide no stable positive risk-level response.

Construction choices nevertheless affect measured responses. At $h = 3$, replacing month-end log squared midquotes with the log monthly mean of squared daily midquotes changes the six- and twelve-month coefficients by -0.0164 and -0.0163, with simultaneous intervals of $[-0.0326, -0.0002]$ and $[-0.0287, -0.0039]$. The underlying level responses remain imprecise. A last-quote six-minus-one-month contrast also narrowly excludes zero at one bandwidth but includes zero at the other. These are differences between construction methods; they say nothing about whether terminal variance rose. Additional daily quote regressions at separations of 0, 1, 5 and 20 quoted days have simultaneous intervals containing zero.

8.3 Distributional shape, denomination, and predictive content

Other exploratory results describe the shape of the response. At the $t + 3$ endpoint, the amplitude of downside moves rises relative to upside moves: the centered downside-minus-upside

amplitude coefficient is 0.590 normalized percentage points, with an eight-margin simultaneous interval of $[0.084, 1.097]$ (Appendix Figure B.1). The total downside partial moment and the paired conditional-variance response remain imprecise, so this is a shift in the relative size of losses and gains, not an increase in overall dispersion.

On 252 IV-date-aligned origins, the response of basket-relative minus USD log realized path risk is -0.0408, with interval $[-0.0787, -0.0030]$ (Appendix Table B.2). Neither individual risk coefficient excludes zero. The ratio therefore reflects denomination and comovement with the basket currencies, not higher absolute peso risk.

Market risk measures can also be informative predictors without responding precisely to TPU. Across 189 recursive evaluation origins, adding IV reduces mean absolute log-risk error from 0.4256 to 0.3900; the paired interval for the change excludes zero. Squared-error and retransformed variance-level loss improvements are imprecise. On 134 common origins, adding the previous month's recursively constructed TPU innovation to the IV rule changes absolute and squared losses by $+0.0009$ and $+0.0041$, with both intervals containing zero (Appendix Table C.1). Implied volatility thus helps forecast realized risk even though it does not respond detectably to TPU.

9 Survey expectations and forecast information

9.1 Reported forecasts and matched target prices

The main comparison uses the reciprocal of a point median on 258 matched origins. Additional survey exercises expand the horizon and aggregation choices. Published EEE medians are available for 260 news months per question, with a narrower verified-date sample of 69. Their simultaneous response intervals include zero. We use the microdata to ask whether the reported consensus and the eventual reference price move differently when aligned to the same target. For survey month $m = t + 1$, the three EEE questions refer to target months $t + 3$, $t + 12$, and $t + 24$. Define the reported consensus as $F_{m,k} = 100 n_{m,k}^{-1} \sum_i \log F_{i,m,k}$ over positive CLP/USD responses, and let $R_{m,k} = 100 \log E_{m+k}$ denote the corresponding target-month reference price. Their difference $A_{m,k} = R_{m,k} - F_{m,k}$ is positive when the realized reference peso is weaker than the geometric survey consensus. It is a calendar-matched forecast error, rather than an excess return.

Appendix Table D.1 reports projections of the consensus, target price, and error on the original standardized innovation, baseline controls, and the same question's consensus from survey month $m - 2$. Each question has 260 news months, with all three outcomes using identical rows and controls. Consequently, the error coefficient equals the target-price coefficient minus the consensus coefficient. The three error responses are 0.542, 0.829, and 0.998 log percentage points. Their nine-margin simultaneous intervals, jointly accounting for the three outcomes and three questions, all include zero under both reported bandwidths. We find no evidence that TPU innovations predict systematic forecast errors, although the intervals still admit errors as large as 1.5–3.4 log percentage points.

Aggregation matters. A geometric consensus is not an arithmetic expectation of the peso's dollar value, and taking reciprocals before averaging gives large weight to a few very small reported quotations, which we retain. The historical microdata contain many responses but only 260 distinct news months per question, and they do not reproduce each month's first public release.

9.2 Forecast performance against dated price rules

The two-month question permits a common-target comparison with the three-month terminal outcome. Both the survey forecast and realized dollar value are normalized by the original $t - 1$

reference quote. A flat-price benchmark uses the latest reference rate whose legal effective date precedes the survey month's first day, and holds that price fixed through the target month.

Appendix Table D.2 shows that the survey improves continuous forecast losses over this benchmark in 219 common months. Mean absolute error falls from 4.695 to 4.367 percentage points, and mean squared error from 38.234 to 32.901 squared percentage points. Paired two-loss simultaneous intervals exclude zero at both Bartlett bandwidths. The directional hit-rate advantage is much smaller: 71.69% versus 69.86%, with an interval that includes zero. Forecast magnitude and directional accuracy thus yield different assessments.

A further comparison uses a recursively estimated price rule with an intercept and three dated monthly changes in dollar-per-peso prices. Training uses only earlier origins with resolved target dates and requires at least 60 observations. Across 156 common evaluation months, the survey has lower absolute and squared losses; the respective differences are -0.462 and -6.885, with simultaneous intervals below zero at both bandwidths. The survey therefore contains predictive information beyond these price rules. That advantage does not vary detectably with TPU: projections of survey-minus-flat losses on TPU innovations have simultaneous intervals containing zero. Two caveats apply. We observe legal effective dates, not exact publication or response times, and because we examined these data repeatedly, the comparison is not a blind holdout test.

10 Expected excess returns

Testing H3 requires investable returns measured over common cash-flow dates. Currency appreciation alone omits the local asset payoff and funding cost, and policy rates, auction yields, or indicative forward marks cannot reconstruct executable returns without instrument prices, settlement terms, and transaction costs. Our search of candidate sources, including CME peso futures and Chilean clearing prices, did not yield a consistent historical return series. H3 therefore remains untested: the price, risk, and survey results cannot show that investors received higher expected compensation for bearing peso risk.

11 Conclusion

Point estimates suggest that U.S. TPU innovations are followed by peso weakness, but not in the pattern a risk-compensation channel predicts. The news-month response is near zero, and cumulative depreciation reaches 0.63 log percentage points only over the following three months. This delayed response is imprecisely estimated, concentrated in 2018–2019, and sensitive to the choice of news measure. Because the model's prediction concerns the current price, it does not confirm the mechanism.

The joint test points the same way. On 258 common news months, neither current depreciation nor the log variance of terminal dollar value shows the required positive response, and survey-implied forecast changes are small. The variance interval is wide, however, admitting increases of up to about 45%, so the data cannot rule out sizable risk effects.

Implied volatility and survey forecasts are informative in their own right: implied volatility improves forecasts of realized risk, and the survey outperforms simple price rules. Neither responds detectably to TPU. The compensation prediction itself remains untested for lack of matched investable returns. Distinguishing among channels more sharply will require variation that separates uncertainty from news about fundamentals, prices and expectations synchronized with the relevant information set, and independent data on investor exposures and executable returns.

A Baseline specification checks

Table A.1 reports the baseline responses and Table A.2 the sensitivity checks. With identical controls, changes in the first-stage coefficient vector alone leave the downstream linear price and risk coefficients unchanged at a fixed innovation scale. Alternatives that change controls, outcomes, sample or residual scale need not preserve the estimand. Calendar-month indicators and a set of more recent economic-period global controls yield $h = 3$ point estimates of 0.663 and 0.621 on the original scale. The intervals for their direct changes relative to the baseline include zero. These comparisons use economic dates; we have not reconstructed historical release dates for all controls.

Table A.1: Baseline responses to a one-standard-deviation TPU innovation

h	N	Estimate	95% HAC CI	HAC p	Block 12 CI	Block 24 CI
A. Cumulative depreciation (log percentage points)						
0	261	−0.016	[−0.394, 0.361]	0.933	[−0.452, 0.420]	[−0.499, 0.467]
1	260	0.165	[−0.279, 0.608]	0.465	[−0.369, 0.698]	[−0.424, 0.753]
3	258	0.626	[−0.035, 1.288]	0.063	[−0.211, 1.464]	[−0.237, 1.490]
B. Log reference-rate realized variance						
0	261	−0.045	[−0.106, 0.015]	0.143	[−0.128, 0.038]	[−0.131, 0.041]
1	260	−0.022	[−0.092, 0.047]	0.526	[−0.107, 0.062]	[−0.109, 0.064]
3	258	0.068	[−0.026, 0.162]	0.155	[−0.020, 0.156]	[−0.019, 0.155]

Notes: HAC intervals are pointwise. Each block column reports 95% simultaneous intervals across the three primary horizons within each panel. The bootstrap uses 9,999 circular-block draws at each block length and reestimates the TPU projection and residual standard deviation.

Table A.2: Baseline sensitivity: three-month depreciation and next-month risk

Specification	N	$\widehat{\beta}_3$	p	N	$\widehat{\theta}_1$	p
Baseline	258	0.626	0.063	260	−0.022	0.526
Six dynamic lags	258	0.588	0.107	260	−0.019	0.600
Reduced global controls	258	0.628	0.062	260	−0.022	0.543
Monthly-average reference rate	258	0.554	0.070	260	−0.023	0.512
log(TPU)	258	0.623	0.066	260	−0.022	0.529
Sample extended through July 2026	280	0.628	0.052	282	−0.031	0.334
HAC bandwidth 6	258	0.626	0.059	260	−0.022	0.512
HAC bandwidth 12	258	0.626	0.051	260	−0.022	0.513
Omit global financial crisis	245	0.595	0.085	249	−0.026	0.467
Omit 2018–2019 trade-war window	231	0.203	0.602	235	0.017	0.633
Omit early COVID-19 window	245	0.654	0.066	249	−0.022	0.533
Omit Chilean stress window	250	0.614	0.081	254	−0.014	0.700

Notes: Two-sided pointwise HAC p-values. The episode exclusions remove outcome windows intersecting September 2008–June 2009, January 2018–December 2019, March–December 2020, or October 2019–February 2020. When controls, sample or transformations change, the innovation is reestimated.

B Supplementary terminal and currency-risk evidence

The four-horizon variance exercise in Table B.1 uses an eight-margin family that pairs current depreciation with terminal variance at each horizon. Its intervals differ from the four-margin 258-

origin family in the main text. For $h = 3$, the relevant intervals are those in Table 1.

Table B.1: Current depreciation and conditional variance of terminal dollar value

h	N	Current depreciation	Terminal variance (% change)
1	260	−0.012 [−0.539, 0.515]	20.40 [−21.02, 83.54]
3	258	−0.012 [−0.539, 0.514]	2.08 [−30.16, 49.21]
6	255	0.005 [−0.529, 0.539]	−6.81 [−37.72, 39.44]
12	249	−0.035 [−0.571, 0.501]	3.83 [−35.91, 68.21]

Notes: Current depreciation is in log percentage points and uses each terminal horizon's sample. Variance coefficients come from a Gaussian working likelihood and are transformed as $100(\exp(\widehat{\theta}_h) - 1)$. The eight-margin 95% simultaneous intervals use 999 common 12-month circular-block draws.

For normalized terminal dollar-value change $Y = 100(Q_{t+h}/Q_{t-1} - 1)$, define loss $L = \max(-Y, 0)$, gain $G = \max(Y, 0)$ and $D = 1(Y < 0)$. The side-specific mean magnitudes $\bar{a}_-$ and $\bar{a}_+$ give a centered amplitude contrast

$$C = (L - \bar{a}_- D) - (G - \bar{a}_+(1 - D)).$$

Its complement in $L - G$ is the frequency contribution $\bar{a}_- D - \bar{a}_+(1 - D)$. The estimates retain all origins and recompute both anchors within each common bootstrap path.

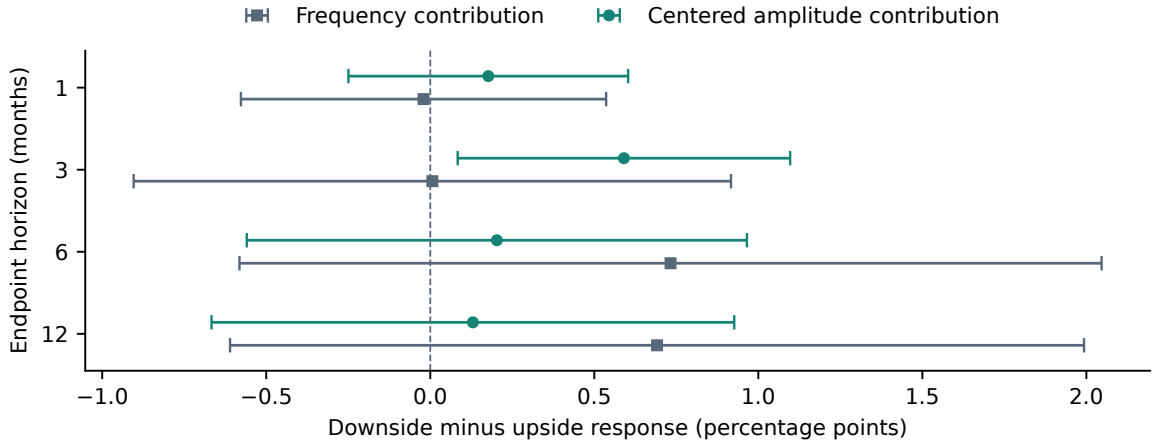


Figure B.1: Frequency and centered-amplitude contributions to downside minus upside

Notes: Four terminal horizons and two contributions form one eight-margin family. Units are normalized dollar-value percentage points per original TPU standard deviation. Intervals use 999 common 12-month circular-block draws, reestimating the innovation and both empirical anchors.

The relative-risk comparison uses common daily intervals for the peso's dollar return and its return relative to an equal-weight basket of AUD, CAD, NZD, BRL and MXN dollar log returns. The log-risk difference equals the log ratio of their realized sums of squares. Its sample and target differ from the terminal-payoff main table.

Table B.2: Price and realized risk on IV-date-aligned paths

Outcome	Estimate	95% simultaneous CI	Family p
Current depreciation	−0.0422	[−0.5422, 0.4579]	0.999
Cumulative depreciation to $t+3$	0.6411	[−0.2204, 1.5025]	0.227
USD log realized risk	0.0234	[−0.0387, 0.0854]	0.828
Basket-relative log realized risk	−0.0175	[−0.0897, 0.0548]	0.961
Cumulative minus current depreciation	0.6832	[−0.2162, 1.5827]	0.213
Basket-relative minus USD log risk	−0.0408	[−0.0787, −0.0030]	0.030

Notes: All outcomes use the same 252 news-month origins. Price outcomes are in log percentage points; risk outcomes are log annualized realized variance. The six-margin family uses 999 common 12-month circular-block draws, reestimating the baseline TPU projection and normalization.

C Supplementary risk forecasting

Both rules in Table C.1 train on at least 60 earlier completed targets whose reference-price effective dates precede the evaluation IV quote date. These date restrictions do not guarantee that quotes and public information were available in real time. The log-risk results, evaluated on 189 origins, and the incremental-TPU comparison on 134 origins each retain their own inference family.

Table C.1: Recursive forecast comparisons for realized USD log risk

Loss	N	Restricted	Augmented	Difference	95% simultaneous CI
A. Add implied variance to the realized-risk rule					
MAE	189	0.4256	0.3900	−0.0356	[−0.0622, −0.0089]
MSE	189	0.3156	0.2783	−0.0372	[−0.0755, 0.0010]
B. Add lagged TPU to the IV rule on common support					
MAE	134	0.3217	0.3226	0.0009	[−0.0129, 0.0148]
MSE	134	0.1990	0.2031	0.0041	[−0.0103, 0.0185]

Notes: Difference is augmented minus restricted. Panel A uses a two-loss family; panel B reports the two TPU margins of a four-margin family that also includes adding IV. Intervals use paired loss differences and Bartlett bandwidth 12. The two panels have different evaluation samples.

D Supplementary survey responses and point forecasts

Table D.1 uses the mean log positive individual response rather than the reciprocal median used in the main joint table. Its matched forecast-error identity holds within this alternative aggregation, while cross-table differences also reflect different transformations, dates and controls. Table D.2 evaluates point predictions against historical price rules on each panel's own common support.

Table D.1: Responses of reported forecasts and realized prices at matched targets

Survey h	News h	N	Reported	Realized	Error [95% CI]
2	3	260	0.083	0.625	0.542 [−0.398, 1.482]
11	12	260	0.017	0.847	0.829 [−0.440, 2.099]
23	24	260	0.039	1.036	0.998 [−1.442, 3.438]

Notes: Outcomes are in log percentage points per baseline TPU standard deviation. The forecast is the monthly mean log positive response and the error equals realized minus reported. The 95% intervals shown are from the nine-margin joint family with Bartlett bandwidth 24. All nine simultaneous intervals include zero.

Table D.2: Survey point forecasts compared with historical price rules

Loss	N	EEE	Benchmark	Difference	95% simultaneous CI
A. Latest applicable pre-survey reference price					
MAE	219	4.367	4.695	−0.327	[−0.614, −0.040]
MSE	219	32.901	38.234	−5.333	[−8.883, −1.783]
B. Recursive three-lag price rule					
MAE	156	4.014	4.476	−0.462	[−0.837, −0.087]
MSE	156	25.534	32.419	−6.885	[−12.009, −1.761]

Notes: Difference is EEE minus benchmark. MAE is in percentage points of simple dollar value change; MSE is in squared percentage points. Each panel uses paired loss differences and a two-loss HAC family (Bartlett bandwidth 12). The evaluation samples differ across panels.

E Supplementary volatility-tenor and risk-reversal quotes

Table E.1: Supplementary Bloomberg option-quote responses

Indicator	h	N	Coefficient	95% simultaneous CI
A. Log squared midpoint quotes (42-margin curve family)				
1M	0	261	−0.0235	[−0.0967, 0.0497]
1M	1	260	−0.0231	[−0.0904, 0.0443]
1M	3	258	0.0275	[−0.0342, 0.0893]
2M	0	261	−0.0232	[−0.0884, 0.0420]
2M	1	260	−0.0215	[−0.0807, 0.0376]
2M	3	258	0.0233	[−0.0335, 0.0802]
6M	0	261	−0.0251	[−0.0821, 0.0319]
6M	1	260	−0.0267	[−0.0796, 0.0262]
6M	3	258	0.0103	[−0.0444, 0.0650]
12M	0	261	−0.0214	[−0.0761, 0.0332]
12M	1	260	−0.0277	[−0.0785, 0.0231]
12M	3	258	0.0048	[−0.0523, 0.0619]
B. Signed three-month risk reversal (12-margin family)				
RR midpoint	0	251	−0.0145	[−0.1598, 0.1308]
RR midpoint	1	249	−0.0341	[−0.1515, 0.0832]
RR midpoint	3	245	−0.0217	[−0.1345, 0.0911]

Notes: The original 261-month TPU normalization is fixed. Each panel is a separate family with Bartlett bandwidth 12; the curve family also includes last-quote levels and tenor contrasts, and the RR family includes last quotes, spreads and same-date LSEG IV. Only the level midpoint rows are displayed. RR units and sign follow the supplier. These rolling quote levels are supplementary risk indicators and do not enter the four-margin main criterion.

F Supplementary timing and exposure checks

Announcement windows. A descriptive check pairs the last common pre-event and first common post-event IV and spot date labels for 15 official trade announcements. The comparison measures subsequent realized path risk relative to a pre-event window using 20 common quote intervals on each side, excluding the event interval itself. For both LSEG and reference spot series, IV and the realized risk ratio move in the same direction in 5 cases and in opposite directions in 10. The data do not record exact announcement or quote times, 13 future windows overlap another event's future window, and the realized window is shorter than the IV tenor. These comparisons are descriptive; they do not identify announcement effects or test option forecast calibration.

Exposure proxies. Across six currencies, a gradient based on the 2000–2002 average share of goods exports destined for the United States has a three-month coefficient of −0.004 log percentage points per 10 percentage points of export share; the simultaneous interval includes zero. In the Chilean analysis, the exposure proxy is the net nonresident foreign-currency derivative position relative to total notional positions, lagged by two calendar months. Over March 2007–June 2024, the interaction with a one-standard-deviation increase in this state changes the current and cumulative depreciation responses by 0.187 and −0.119 log percentage points, respectively. Their simultaneous intervals are $[-0.787, 1.160]$ and $[-1.156, 0.917]$. The corresponding interaction intervals also include zero for USD log risk and the relative risk ratio. These exercises vary observed proxies, whose coverage and construction differ from the net peso exposure of the modeled investors.

G Reading and reproducing the estimates

The accompanying source package contains the editable tables and vector figures, selected estimation summaries, and an index mapping those summaries to their source files. The reported estimates were produced with Stata/MP 18. In the research workspace, the baseline entry point is `stata/run_all.do`; the higher-draw bootstrap is produced by `stata/23_high_precision_bootstrap.do`. Later analyses use their own samples and inference families. The package is sufficient to compile this manuscript; the underlying historical quote exports, individual survey responses, and full estimation archive are separate.

The baseline bootstrap and the later exercises address dependence differently. The former resamples month-indexed tuples with circular blocks and keeps aligned horizons intact. Some extensions use calendar-indexed score covariance and shared multipliers; forecast comparisons use paired loss differences and HAC. The confidence level applies to the family described in each table. As noted in Section 5, most extensions are exploratory. In a simplified fitted Gaussian process with a dominant root near 0.993, nominal 95% bootstrap bands cover the six true responses on about 87% and 83% of 199 simulated paths for blocks of 12 and 24 months, respectively. Because the simulated process is simplified, these rates indicate rather than measure the coverage of the empirical baseline.

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